

First Officer / Pilot Monitoring Documentation

Document info:

Doc. Revision: Rev 2

Plugin Version: V1.2 On Hand Checklist

Hello there, this document will show you how to use the FO/PM plugin and how to interact with the UI.

¡DISCLAIMER!: This plugin is developed for the Toliss A319, A320, A20N, A321 and A21N only (For now). You need to have the latest version of FlyWithLua installed on your sim, and the aircraft updated to A319 V1.12.1, A320/A20N V1.3.2 or A321/A21N V1.9.1. For installation instructions and file configuration please read the "README" file, this file is only for explain how to use the plugin.

///// Commands /////

There are 6 main commands to interact with the FO, you need to assign a key from your keyboard and you can find them in the Xplane's settings menu/Keyboard/Toliss_A325_FO

- Confirm validation for checklist (Response Check)
- Command Flaps 1 Position Down
- Command Flaps 1 Position Up
- Command Gear Down
- Command Gear Up
- Toggle FO interface (Show/Hide)

///// UI Presentation /////

The UI is a Float Window that appears in your sim when it's open by the command or macro.

In the UI you will find 3 main pages: "Main", "Settings" and "Briefing", all 3 pages are divided in sections separated by a grey line.



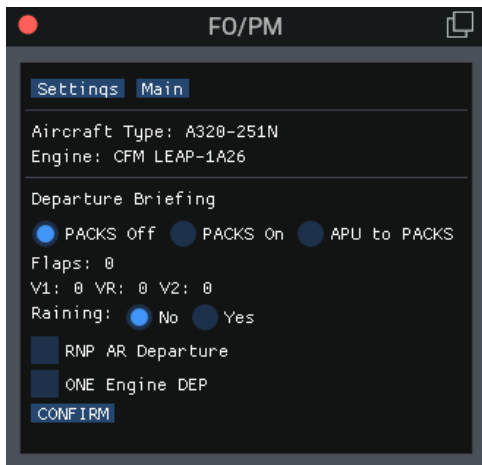
Main Page:

1st Section: Buttons to change between pages, and there is an X square for check the checklist items to advance.

2nd Section: Current flight phase, and the RECOVERY button when a saved flight is available.

3rd Section: Available Checklist. While a checklist is running it shows "Executing Checklist", or the current item and its answer if "Interface Checklist" is on.

4rd Section: Available Procedures

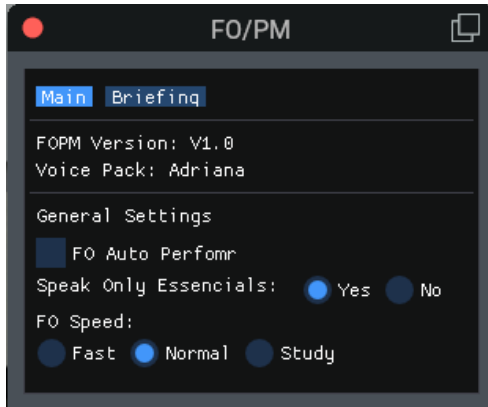


Briefing page:

1st Section: Buttons to change between pages

2nd Section: ACF info and live flight data read from the MCDU (Only on departure)

3rd Section: Available briefing config, CONFIRM / DEP CHANGE and WX REQUEST buttons



Settings page:

1st Section: Buttons to change between pages

2nd Section: Logbook (total flight hours and total flights), plugin version, loaded voice pack, procedures pack and checklists pack names

3rd Section: Settings config and the Change PROC/CKLT Pack button

///// UI Features /////

You can move and adjust the window at your discretion, the system will recognize the current position to adjust the size automatically when you change between the pages, and it will keep the size you gave it.

The position is saved every time you close the window, no matter if you close it with the command, the macro or the window's own close button. The window always opens again on the "Main" page. If the saved position is no longer valid, for example because it was on a monitor that is not connected anymore, the window opens in its default position.

///// Controls /////

This is a Phase-Based Plugin, depending on the flight phase, you can see it in the "Main" page, the system will trigger some procedures and activate or deactivate the option to command procedures and checklist. All the available checklist and procedures are controled in the "Main" page by using the buttons to tell the FO to perform them.

The standby phase is the "Preflight", in here the FO will be on standby and will not perform or read any control or situation of the airplane. To initiate or to wake up the FO, you need to perform the "Preliminary Cockpit Prep" procedure.

¡WARNING! If you pushback without the Preliminary Cockpit preparation, the FO will be on Standby all the flight.

The available checklist depends on the flight phase and the previous performed checklist and procedures, all starts via a command button except for one under a specific condition.

• Preliminary Cockpit Prep

Available Checklist:

- Available Checklist:
- Cockpit Preparation checklist
- Before Start checklist
- After Start checklist (the FO launches it himself when performing one engine taxi departure)
- Taxi checklist
- Departure Change checklist (only available after using DEP CHANGE)
- Line Up checklist
- Approach checklist
- Landing checklist
- After Landing checklist
- Parking checklist
- Securing checklist

Once a checklist is completed its button disappears, and it only comes back when the flight needs it again: after a go-around (Approach and Landing), after a rejected takeoff (Line Up) or on the next flight. When the Parking checklist takes you back to "Preflight", every checklist and procedure of the previous flight is reset, so you can fly the next leg without reloading the script.

The available procedures depend on the flight phase and the previous performed procedures and checklists, some are automatic and others are triggered by a command button

Available Procedures:

- Preliminary Cockpit Preparation (Command)
- After Start Procedure (Auto)
- Taxi Procedure (Command)
- Before Takeoff Procedure (Command)
- Enter Runway (Command, hidden while the Taxi or Before Takeoff procedure is running)
- Exit Runway (Command, hidden while the Taxi or Before Takeoff procedure is running)
- Takeoff Procedure (Auto)
- Clean up procedure (Auto if "FO Autoperform" is on)
- 10000ft Climb Procedure (Command / Auto if "FO Autoperform" is on)
- 10000ft Descend Procedure (Command / Auto if "FO Autoperform" is on)
- AP Disengage (Auto when AP is disconnected)
- GO-Around Procedure (Auto)
- After Landing Procedure (Auto)
- Parking Procedure (Auto)

There are two supplementary procedures available:

- One engine taxi departure procedure (Auto, but engine 2 start is triggered by command)
- One engine taxi arrival procedure (Command)

///// Briefing /////

The "Briefing" page is very simple, in the second section are the available configuration for briefing, there are two "sub-pages" depending on the flight phase, departure and arrival briefing, in both you can set the parameters and/or acf configuration for the situation.

On departure, the second section also reads live data from the MCDU and shows your flight (departure, destination and alternate), the takeoff runway, the pitch trim, the thrust setting (TOGA or Flex with its temperature) and the current baro reference. On aircraft with IAE engines it also shows the time elapsed since the last engine shutdown, useful for turnarounds. The takeoff runway shown here is the one the FO reads out in the T.O RWY item of the Line Up checklist ("two seven left"), so open the Briefing page while the captain's MCDU shows the PERF TAKE OFF page to load it.

Once you press CONFIRM on the departure briefing, the button becomes DEP CHANGE. Use it if the departure changes: you can edit the runway, flaps, speeds or packs configuration again, and it will also make the Departure Change checklist available in the Main page. The arrival briefing behaves the same way, its CONFIRM button becomes ARR/APP CHANGE so you can revise the approach type, minimums or autobrakes if the arrival changes.

///// Baro Reference/////

The FO can now set his own baro reference, and there are two ways for him to get the QNH. Which one you use is decided by the "FO Request Weather" setting, explained in the Settings section.

With the setting ON, both briefings show a WX REQUEST button. Pressing it makes the FO request the weather on his own: he navigates the MCDU to the ATSU datalink page, reads your departure, destination and alternate airports from the flight plan, sends the METAR request and waits for the reports to arrive. He then opens the report he needs for the current phase of flight, the departure airport while you are still on the ground and the destination once you are airborne, reads the QNH, calls it out digit by digit and sets his baro reference. He returns the MCDU to the flight plan page when he is done. This uses the aircraft's ACARS datalink, so it needs a Hoppie logon code configured in your ToLiss aircraft.

With the setting OFF, both briefings show a "Set Baro Ref" field instead. Type in the QNH you got from anywhere else and press SET. The SET button only shows up once the value you typed is a pressure the aircraft can actually accept, so if you don't see it yet, check what you entered. From there the FO does exactly the same thing: he calls the value out digit by digit and sets his altimeter, with no Hoppie needed.

Either way the FO switches between hPa and inHg by himself depending on the value, and pulls to STD if you are already above the transition altitude. He says "QNH" before the digits when the value is in hPa, and "ALTIMETER" when it is in inHg. The departure and the arrival briefing each have their own button and field, so you can set a new reference for the destination while in flight.

The Baro Reference item in the Cockpit Preparation and Approach checklists is checked against this QNH, no matter which of the two ways it came from, so it will only pass when the altimeters are actually set to that pressure. If no QNH has been set yet, the item simply checks that both altimeters agree with each other.

///// Settings /////

There are 5 main settings:

- FO Auto Perform (Yes/No)

Enable the FO to autoperform procedures such as gear retraction, Flaps retraction (Clean up procedure), 10000 ft climb and descend procedure (executed at 14000ft by default to comply with high altitude airports).

- FO Request Weather (Yes/No)

Decides how the FO gets the QNH for the baro reference. If yes, he requests the METAR himself through the aircraft's datalink and a WX REQUEST button appears in the briefings; this needs a Hoppie logon code configured in your ToLiss aircraft. If no, the briefings show a "Set Baro Ref" field where you type the QNH yourself and press SET. This setting is off by default.

- Interface Checklist (Yes/No)

If yes, while the FO reads a checklist the Main page shows the current item and its answer, for example "SEAT BELTS... ON", and it moves on at the same pace as his voice. Items whose answer depends on your flight, such as BARO REF, FLAPS SETTING, PITCH TRIM or T.O RWY, show "____" in place of the value. If no, the Main page only shows "Executing Checklist". This setting is off by default.

- Speak Only Essentials (Yes/No)

If yes, the FO will say only the callouts that are mandatory by SOP, and in the checklists he will only answer the essential items (BARO REF, T.O SPEEDS & THRUST, WINDOWS & DOORS, FLIGHT CONTROLS, FLAPS SETTING and T.O RWY), the rest of the items advance silently when you confirm them. If no, the FO will say all the movements and answer every checklist item.

- FO Speed

This is the speed at which the FO will perform the procedures, there are 3 possible speeds:

- Fast: 0.6s between steps.
- Normal: 0.85s between steps.
- Study: 1.1s between steps (Set the "Speak Only Essentials" to "No")

The study speed is designed for pilots who want to understand the flows of the ACF, giving them time to observe the movements of the FO.

All your settings will save automatically every time you change it.

///// Logbook /////

The Settings page keeps a logbook of your flights with the FO: Total Flt Hours and Total Flights. The time starts counting when the aircraft starts moving at pushback and stops when the Parking procedure is completed, which is when the flight is added to the logbook. It is saved in "FO PM/Logbooks/A32S Logbook.lua", so it carries over from one session to the next, and a flight picked back up with RECOVERY keeps counting.

///// Procedures & Checklist Packs /////

The complete set of procedures and checklists can be changed, so you can fly the same aircraft following a different operator's SOP. The plugin includes two packs: Airbus, which are the manufacturer standard procedures, and Avianca 2022, which are airline procedures. The Legacy pack, with the procedures used in V1.0, was retired in V1.2.

To change it, open the Settings page, press "Change PROC/CKLT Pack", select the one you want, press SAVE and reload the script. The names of the loaded packs are shown in the second section

of the Settings page.

///// Limitations /////

In addition to being exclusive to the ToLiss Airbus family, there are three technical limitations you should be aware of:

- **Hopple Logon Code:** The WX Request function uses the aircraft's ACARS datalink to retrieve the METAR, which is provided by the Hopple network. If you do not have a Hopple logon code configured in your ToLiss aircraft, simply leave the "FO Request Weather" setting off and enter the QNH by hand in the briefing instead. Everything else in the plugin works exactly the same either way.
- **A321 Center Tank Fuel Pumps:** Due to a limitation within the ToLiss system logic, the commands for the Center Tank (CTR TK) Fuel Pump Push Buttons are not externally linked/exposed. Consequently, when flying the A321 and A21N variants, the FO will not be able to automatically turn off the center fuel pumps; this action must be done manually by the pilot.
- **External Hardware Conflicts:** If you use physical cockpit hardware (like dedicated landing gear or flaps levers), you might experience synchronization issues where the FO's actions clash with your physical lever positions. Please note that standard flight controls—such as joysticks, thrust levers, and speed brake levers—will function perfectly and do not cause any conflicts.

Thats it, enjoy your flights with your FO.

Best Regards: Admiral Skippie